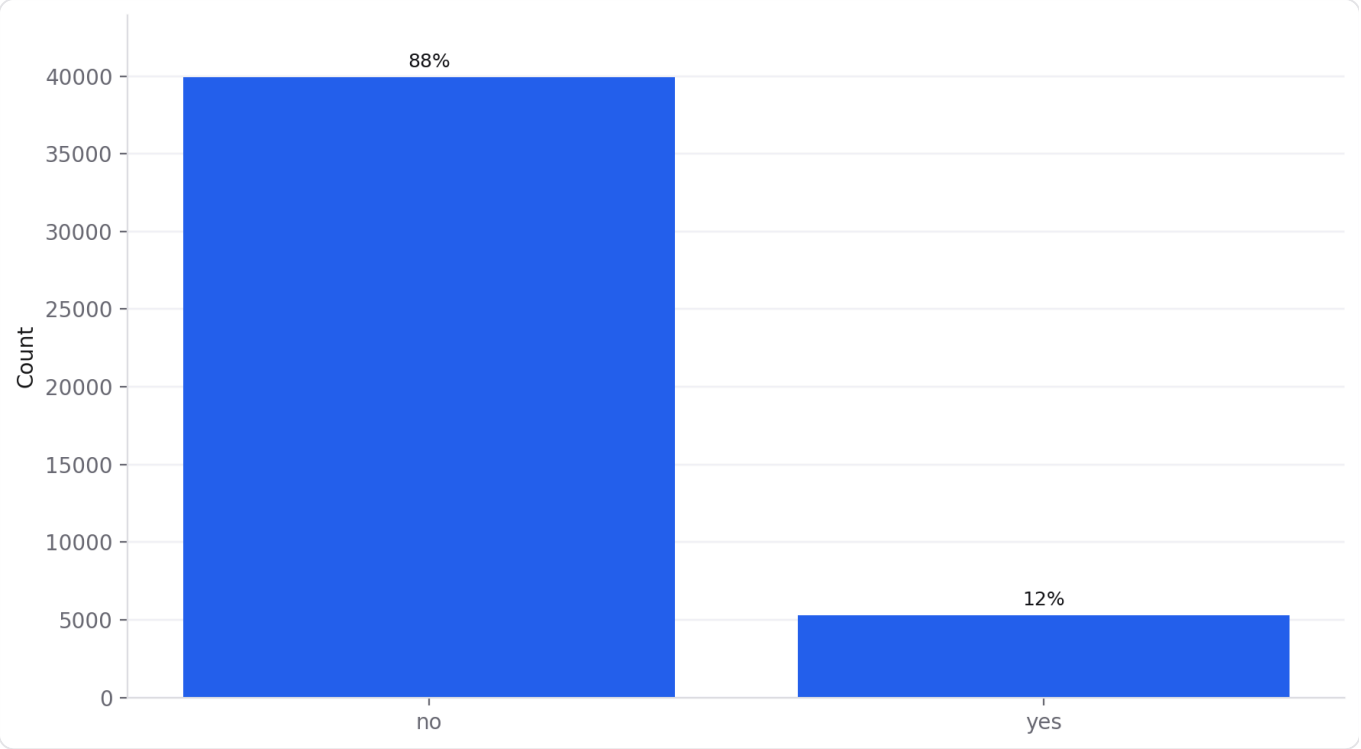
Who will subscribe? An honest model of a bank's phone campaign

Created 3 November 2026 with Data Boss · Data: bank-full.csv

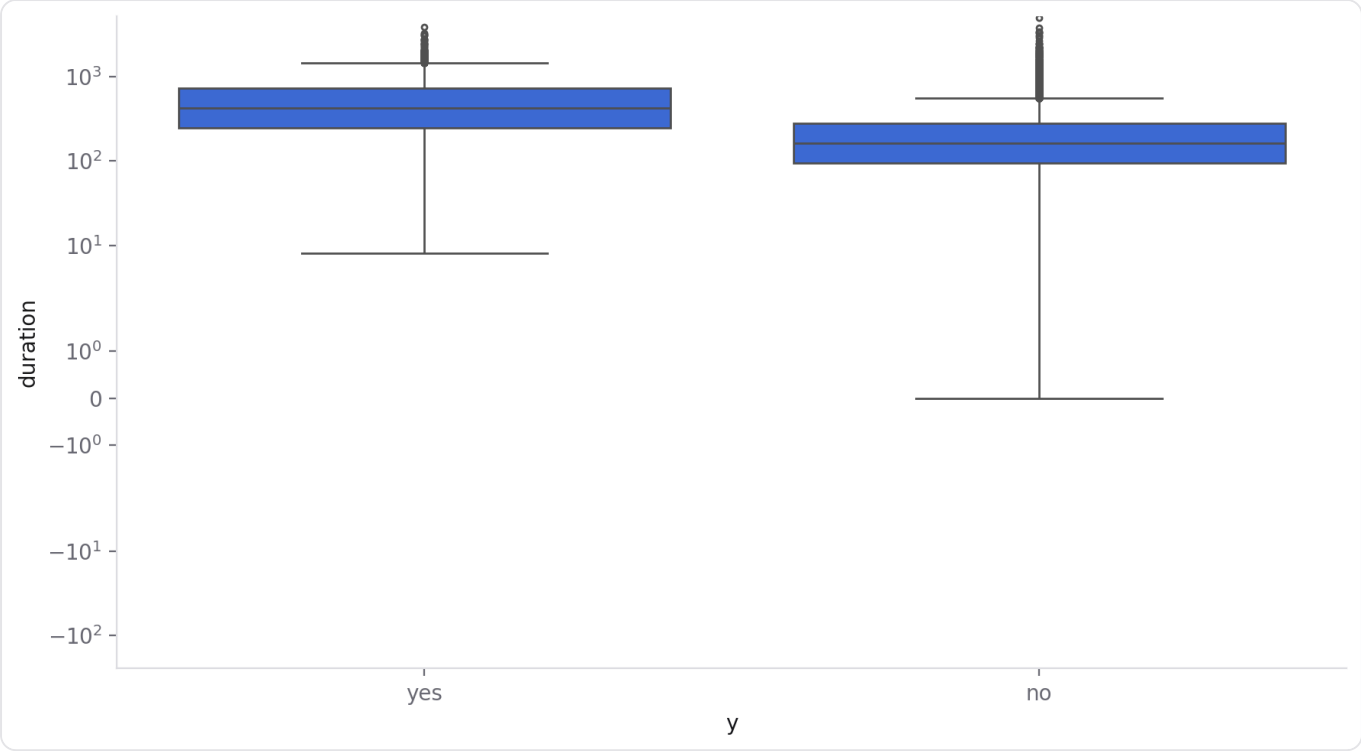
45,211 customers called by a Portuguese bank (UCI Bank Marketing, CC BY 4.0). 11.7% subscribed to a term deposit. The call's duration is only known after the call, so a model that uses it cannot be used to decide whom to call.

How many customers subscribed



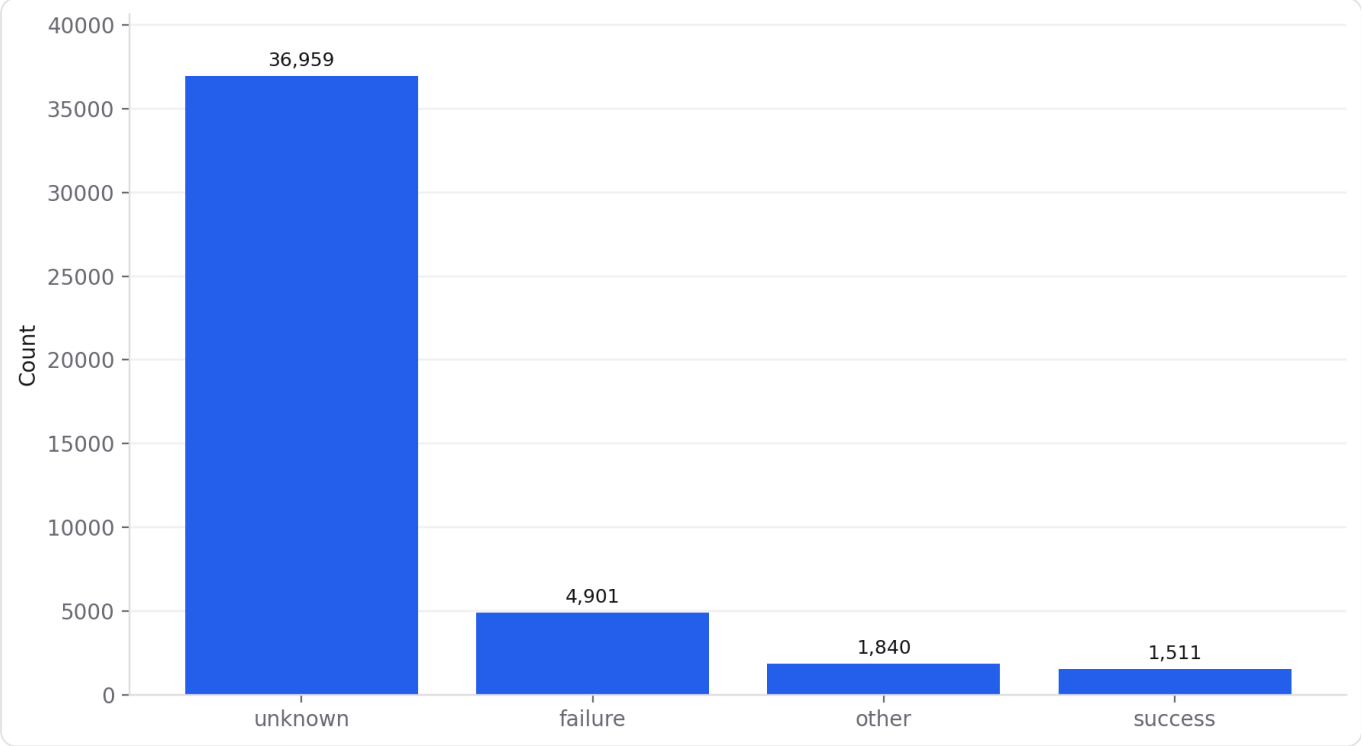
- 2 distinct values across 45,211 rows.
- Most common: no (39,922, 88%).

Call duration (seconds, log scale) by outcome



- Highest median: yes (426). Lowest: no (164).
- Overall median 180; middle 50% between 103 and 319.
- 3,235 point(s) fall outside the whiskers (1.5 × IQR).

Customers by result of the previous campaign



- Highest: unknown (36,959). Lowest: success (1,511).
- unknown alone accounts for 82% of the total.

Model: predicting y (classification)

Model	Accuracy	F1	ROC-AUC	Train F1
Baseline	0.883	0	0.5	0
Linear model	0.9009	0.4503	0.9057	0.4518
Random forest (best)	0.9068	0.5003	0.9306	1

- Random forest predicts y best: 91% of unseen rows right (F1 0.50 for y = yes). Always guessing the most common value would get 88% right.
- Most of the model's skill comes from “duration”. That is fine if it is known at the moment you would make a prediction; if it is only known afterwards (like the length of a sales call before the call), it is data leakage and the score is too optimistic.
- Random forest scores F1 1.00 on the training rows but 0.50 on the test rows: it memorises more than it generalises (overfitting).
- Most important: duration, poutcome, age.
- 36,168 training rows, 9,043 test rows.

Model: predicting y (classification)

Model	Accuracy	F1	ROC-AUC	Train F1
Baseline	0.883	0	0.5	0
Linear model	0.8935	0.2829	0.7718	0.2765
Random forest (best)	0.8949	0.3493	0.7905	1

- Random forest predicts y best: 89% of unseen rows right (F1 0.35 for y = yes). Always guessing the most common value would get 88% right.
- Random forest scores F1 1.00 on the training rows but 0.35 on the test rows: it memorises more than it generalises (overfitting).
- Most important: poutcome, month, housing.
- 36,168 training rows, 9,043 test rows.

Statistics are computed from the data as loaded and cleaned in Data Boss. Associations and differences describe this data; they do not by themselves show cause.